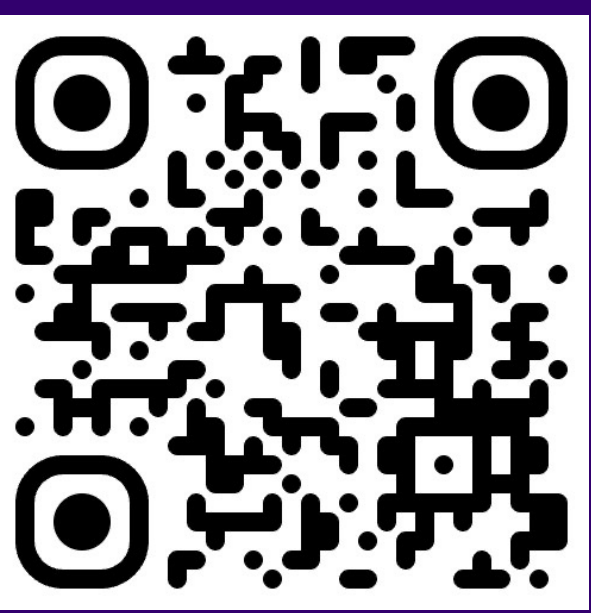
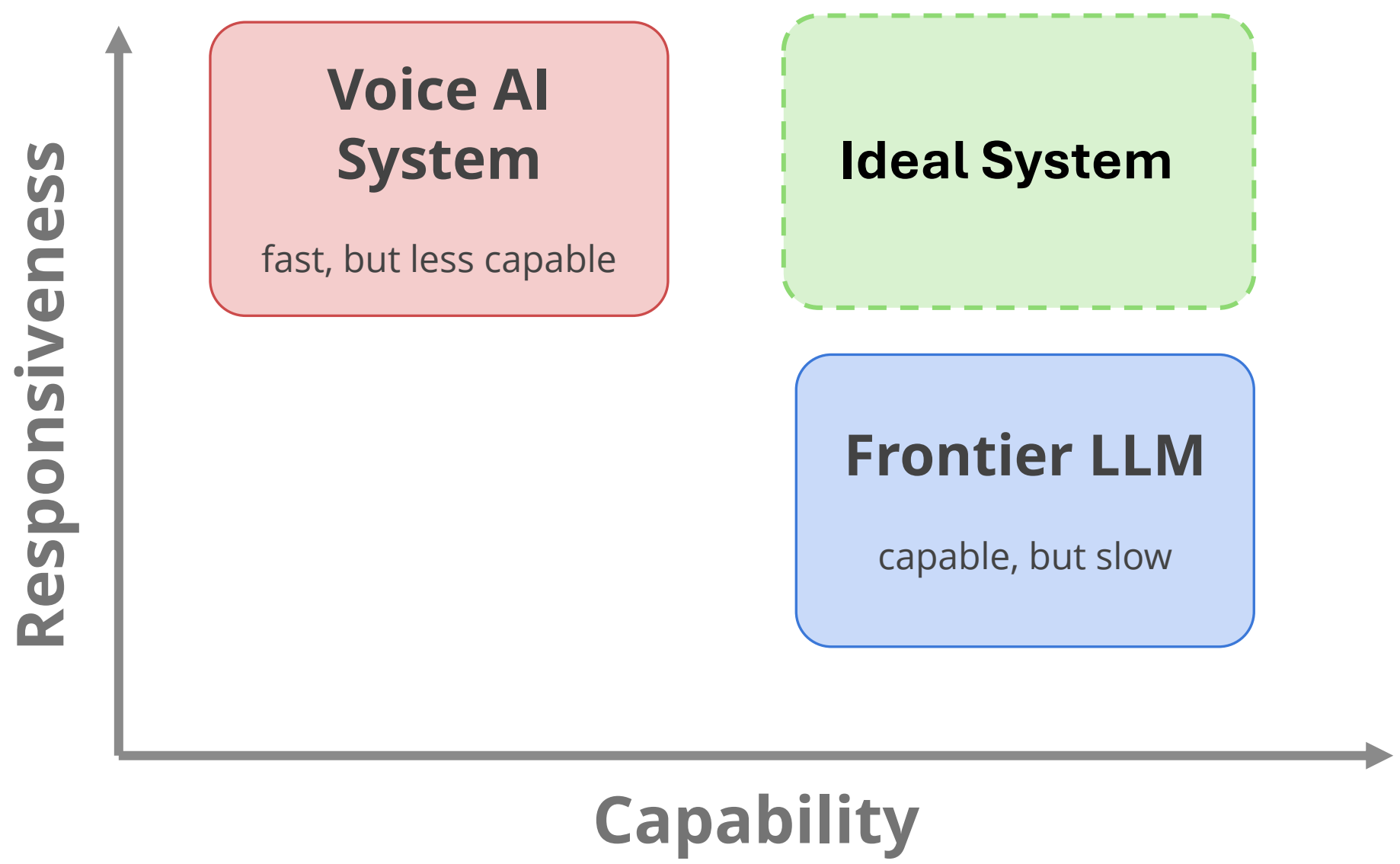




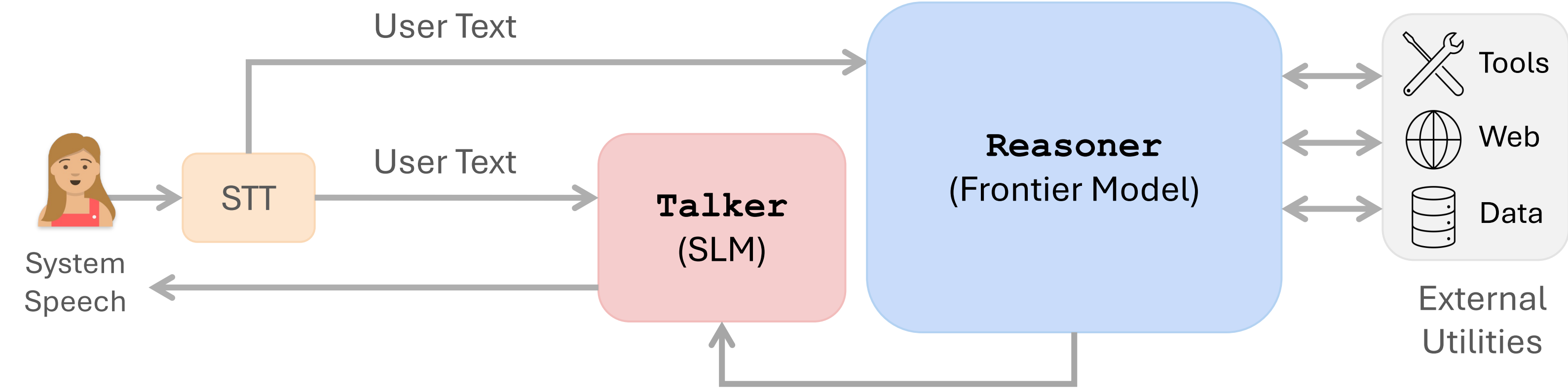
Thinking while Speaking: Inference-Time Knowledge Transfer for Responsive and Intelligent Conversational Voice Agents

Vidya Srinivas*, Zachary Enghardt*, Vikram Iyer, Shwetak Patel
University of Washington

Latency-Capability Tradeoff

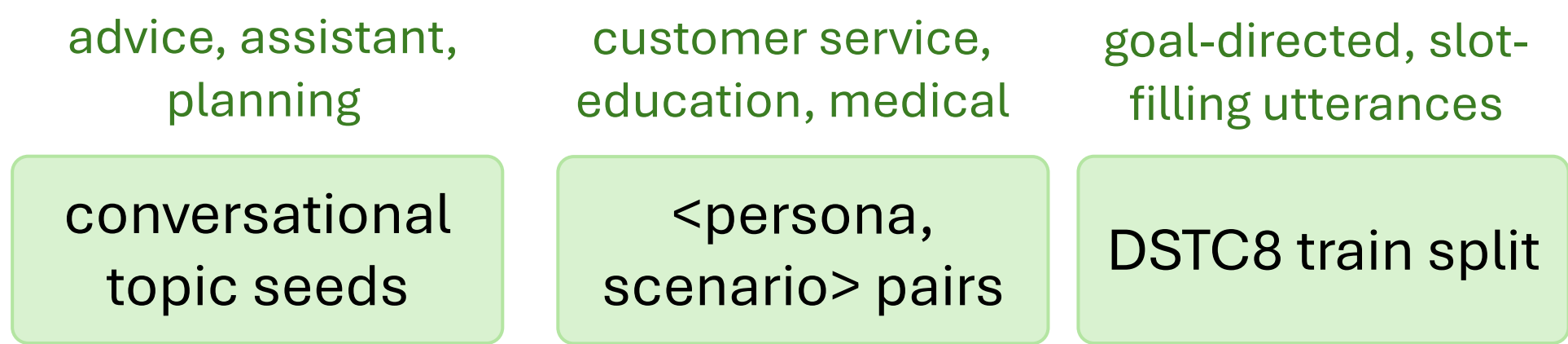


Conversational Infill: Inference Time Collaboration

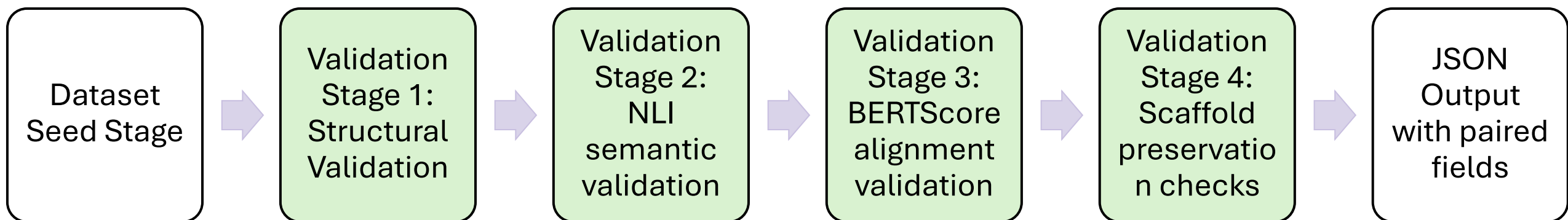


Decoupled, **asynchronous** inference to enable voice models to remain **responsive** but also **capable**.

Seed data: ensuring diversity

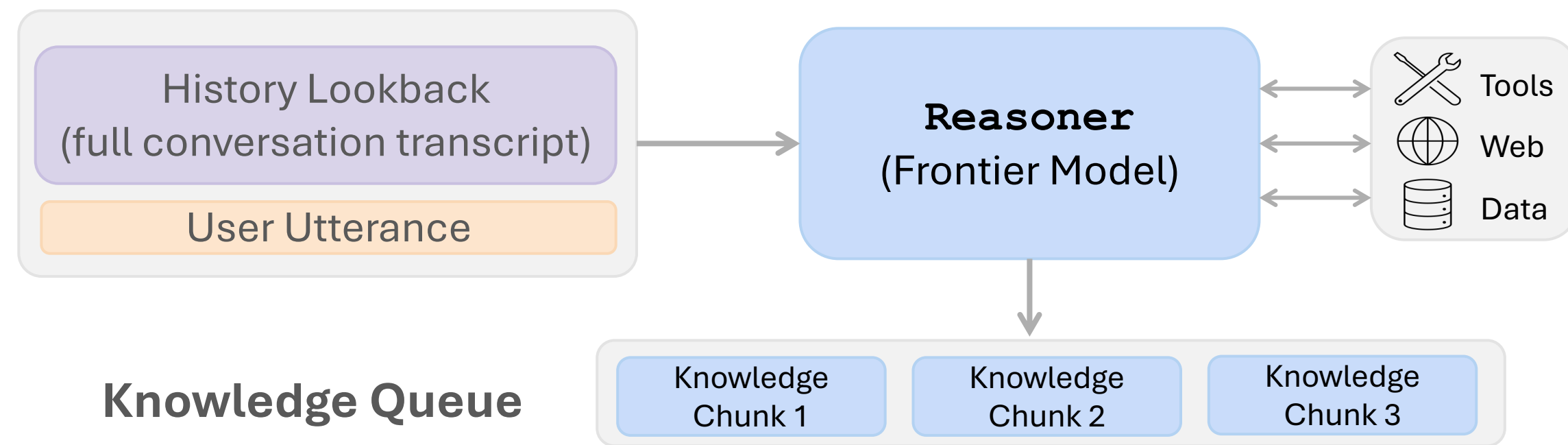


Four-stage data validation protocol

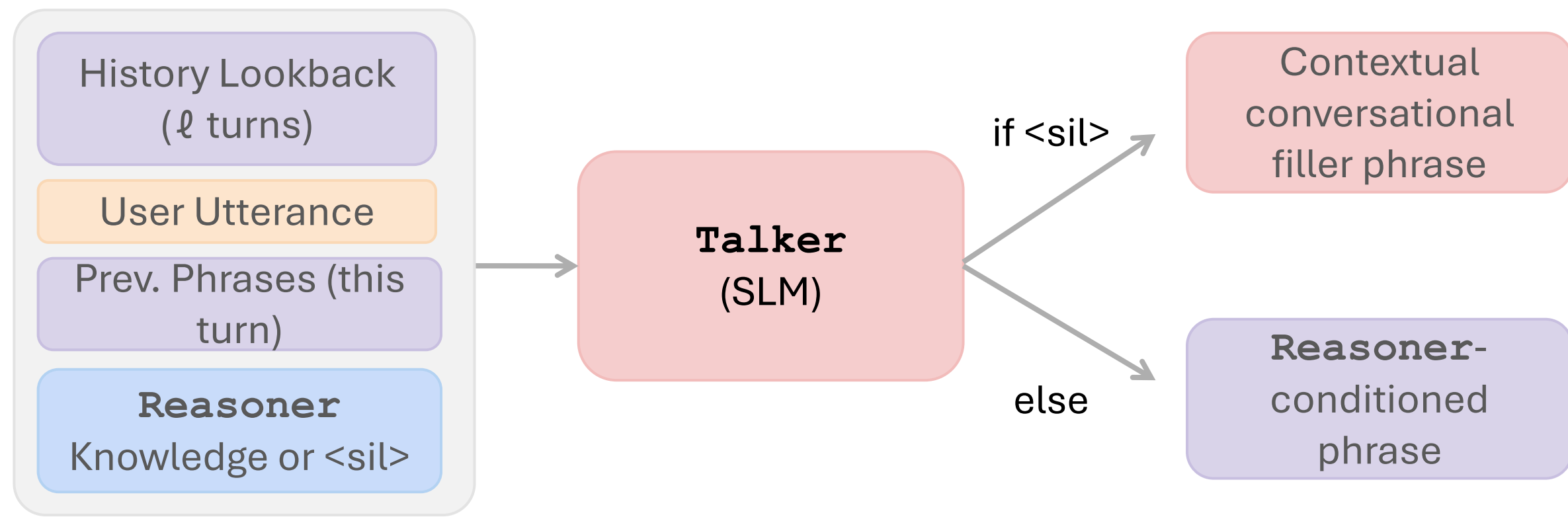


137,111 conversations generated, 8443 (~6.4%) passed validation.
290,571 training examples were generated overall.

Reasoner: the knowledge source

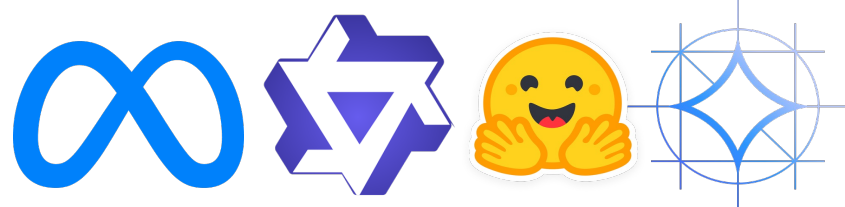


Talker: dynamic information incorporation



ConvFill Configurations

7 finetuned Talker models (135M-1.7B)



3 Reasoner models across families



21 ConvFill configurations evaluated across 4 Talker model families

Evaluation

Validates objective performance via benchmark evals

Demonstrates generalization to tasks **not included in training**: RAG and MCP

User study with end-to-end voice system measures user experience and preferences

Benchmark Evaluations

Single-Turn QA

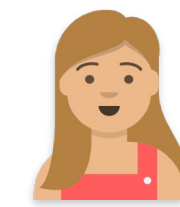
SimpleQA & LLAMA1

Multi-Turn Dialogue

MultiWOZ & EverydayConversations

Evaluated both end-task performance and task adherence across longer context

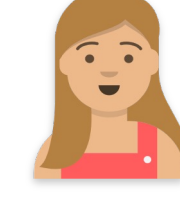
ConvFill: Live, Real-time System



What are some good places in Seattle to get crepes at 4:30 PM today?



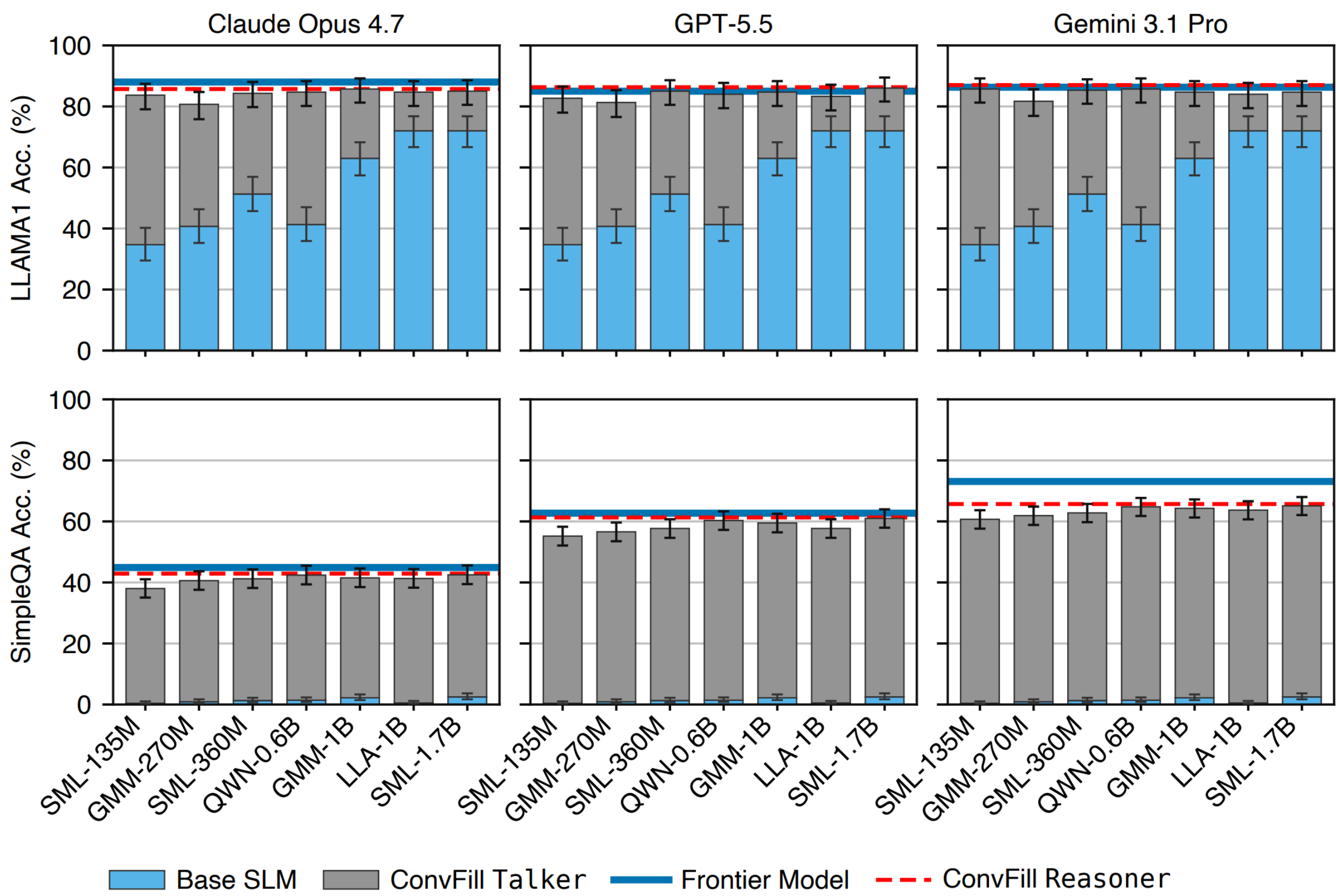
Crepes sound like a great idea! Let me find a place for you. Alright, so ...



What are some good places in Seattle to get crepes at 4:30 PM today?

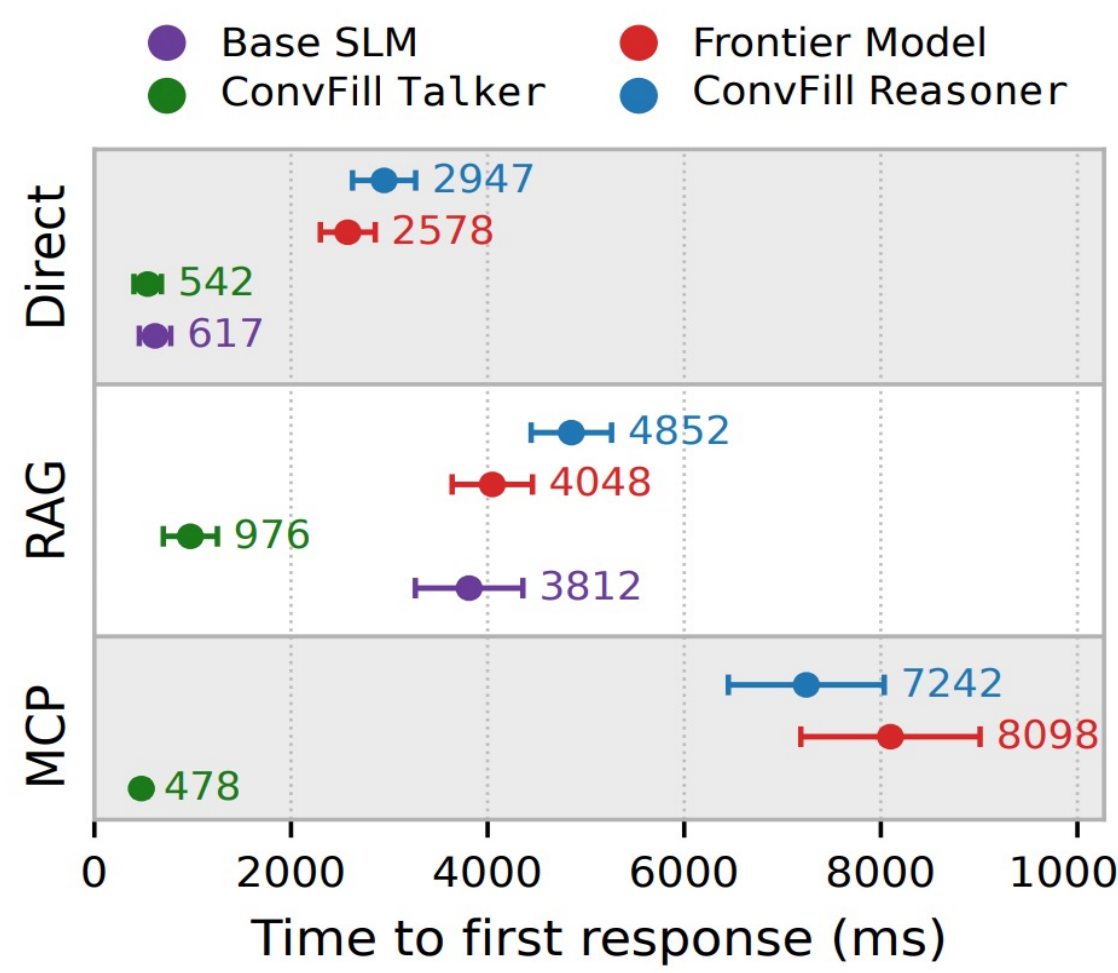
18 participants completed 144 conversations across 3 randomized ConvFill configs

ConvFill Talkers retain Reasoner performance



Talker models close the accuracy gap on single turn QA to **within 6.3%** of Reasoners, improving on base model-only performance by 51%-63%

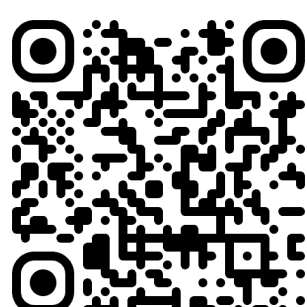
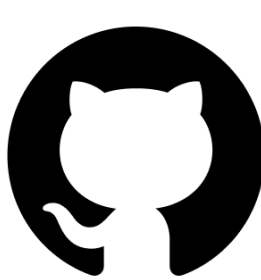
ConvFill preferred for higher-latency tasks



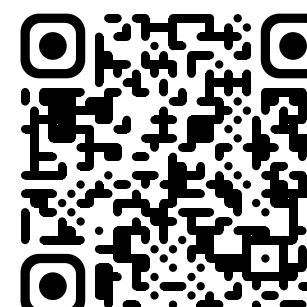
Task	System	Rank 1	Rank 2	Rank 3
Direct	Base SLM	0	3	15
	Frontier	10	7	1
	ConvFill	8	8	2
RAG	Base SLM	1	2	15
	Frontier	5	11	2
	ConvFill	12	5	1
MCP	Frontier	7	11	–
	ConvFill	11	7	–

ConvFill demonstrates architectural flexibility, and presents a promising approach for future voice agents that are both capable and responsive

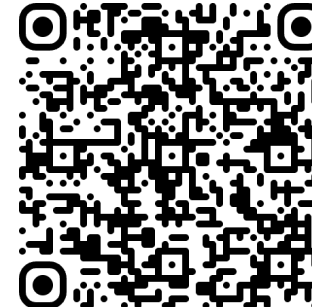
Try it yourself!



Dataset



Models



Demo



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OF COMPUTER SCIENCE & ENGINEERING

